VCCR v0.2.0 — Independent Euler Qualification

SST / A006 • 23 October 2026 • 18 / 18 Python regressions PASS

Result scope: PASS for an exact, dimensionless, unbounded affine Euler benchmark and independently advected material polygons. NOT RUN for a genuine knotted, localized finite-core Euler vortex. Physical SST closure remains UNRESOLVED.

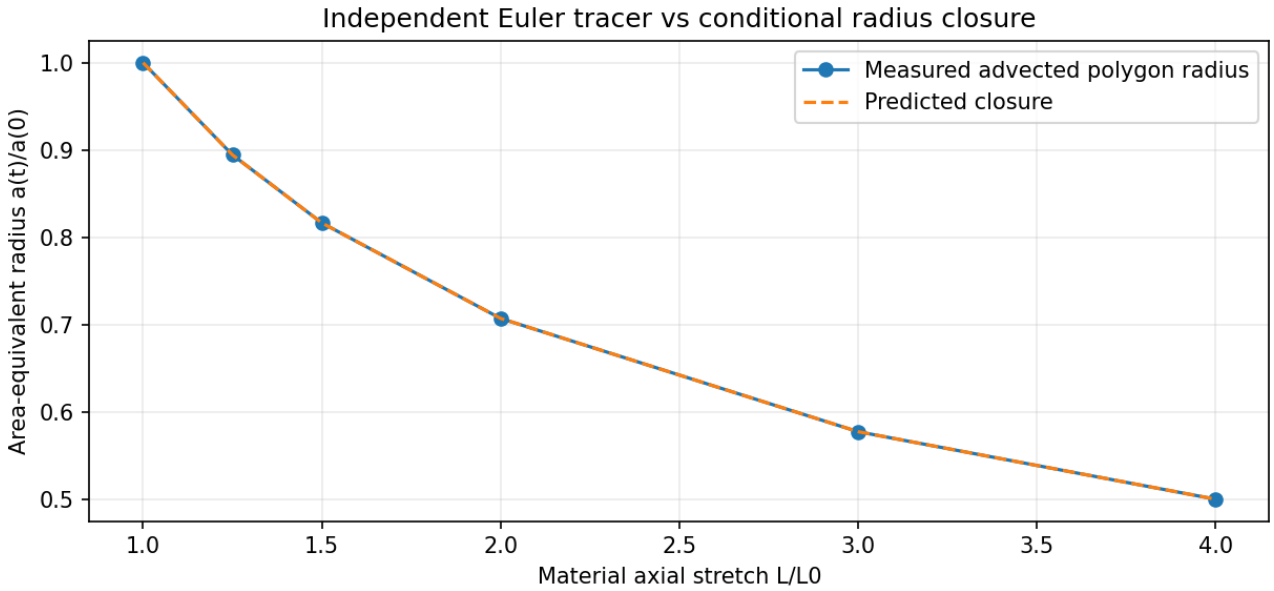
1 Governing equations and independent construction

The flow is $u = (-kx/2 - \Omega(t)y, \Omega(t)x - ky/2, kz)$, with $\Omega(t) = \Omega_0 \exp(kt)$. Its pressure is $p/\rho = -1/2\,[(k^2/4 - \Omega^2)(x^2+y^2) + k^2z^2]$. Direct substitution satisfies incompressible Euler with no external forcing.

Material ring markers are integrated by fourth-order Runge–Kutta (RK4). Polygonal area and $a_{eff} = \sqrt{(A/\pi)}$, axial arclength, circulation $\oint u \cdot dl$, and a finite-difference curl are measured without imposing the radius closure. Numerical probes independently test Euler momentum, divergence and pressure Poisson.

2 Measured circle benchmark

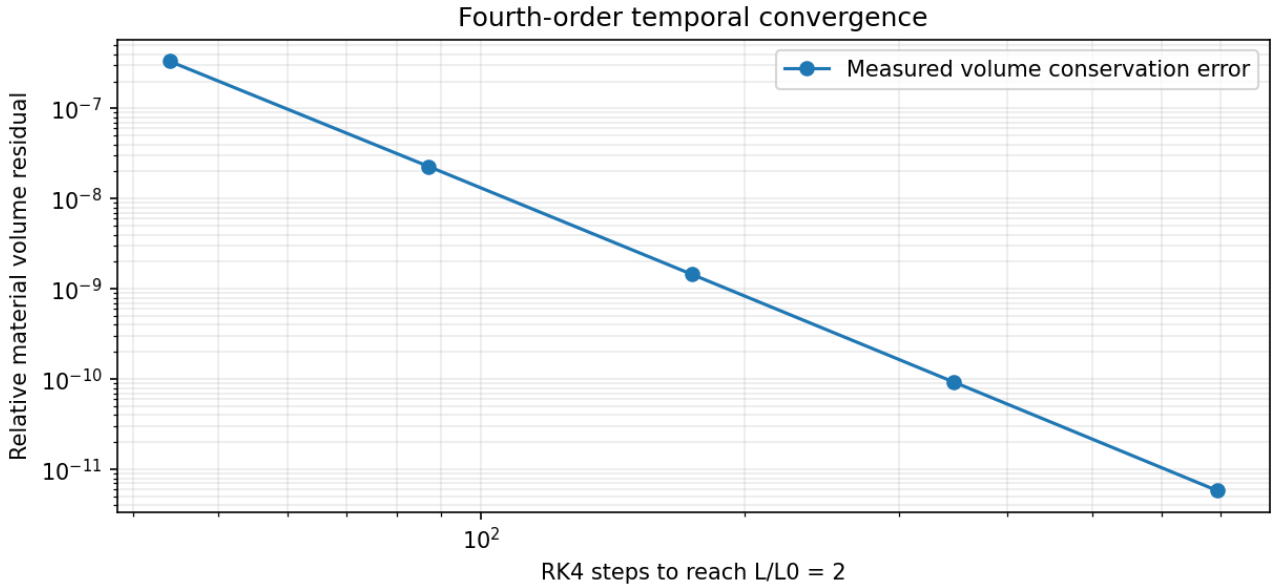
Stretch	Radius measured	Radius expected	Radius rel. error	Volume rel. error	Circulation rel. error
1	1.0000000000	1.0000000000	0.00e+00	0.00e+00	0.00e+00
1.25	0.8944271910	0.8944271910	4.30e-12	8.60e-12	8.63e-12
1.5	0.8164965809	0.8164965809	1.24e-11	2.48e-11	2.49e-11
2	0.7071067812	0.7071067812	4.65e-11	9.30e-11	9.31e-11
3	0.5773502693	0.5773502692	2.49e-10	4.98e-10	4.98e-10
4	0.5000000004	0.5000000000	7.83e-10	1.57e-09	1.57e-09



The two traces overlap at plotted resolution; numerical differences are listed above.

3 Countermodels and numerical convergence

Max sampled Euler momentum residual: 8.04e-11; pressure Poisson residual: 1.06e-09 (dimensionless). Holding Ω constant under nonzero axial stretching is rejected by both Euler momentum and Kelvin circulation checks. The elliptical material section preserves area-equivalent radius while its geometric boundary is not circular.



The time refinement yields an approximately 16-fold error reduction for half the integration step, consistent with fourth-order global integration.

4 Qualification gates

Gate	Result
G0 v010 analytic	PASS (7 legacy tests)
G1 independent exact Euler benchmark	PASS
G2 negative control	PASS
G3 material cross section affine	PASS
G4 native knotted solver	NOT_RUN
G5 independent native vorticity flux	NOT_RUN
G6 spatial temporal convergence native	NOT_RUN
G7 topology and clearance	NOT_RUN
G8 SST clock	NOT_RUN

5 Limits and falsifiable next measurement

The Euler benchmark is unbounded and has uniform vorticity throughout space, not a finite-core tube confined to its material ring. It cannot prove knot stability, geometric ropelength saturation, particle identity or Swirl Clock physics. Separate native-data intake has been implemented, but tested only with labeled synthetic material markers.

Next: supply a source-native Euler or PKLSA finite-core simulation with persistent material IDs, rings along a knotted filament, velocity/pressure samples, independent circulation and vorticity-flux measurements, solver convergence and independent topology/clearance gates. Preserve G4–G8 as NOT RUN until those exist.

References

Chorin & Marsden (1993), A Mathematical Introduction to Fluid Mechanics, 3rd ed., Springer. doi:10.1007/978-1-4612-0883-9.

Majda & Bertozzi (2002), Vorticity and Incompressible Flow, Cambridge University Press. doi:10.1017/CBO9780511613203.

Cantarella, Kusner & Sullivan (2002), On the Minimum Ropelength of Knots and Links, Invent. Math. doi:10.1007/s00222-002-0234-y.